Enterprise Customer Data Platform (CDP)

Architecture, Design Decisions, and Interview Handbook

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Executive Summary

The Customer Data Platform (CDP) is an enterprise-wide customer data management platform designed to create a Single Customer View across all insurance journeys.

The platform consolidates customer information from multiple source systems and maintains:

- Prospect Profiles
- Golden Customer Profiles
- Customer Journey Data
- Customer 360 Views

The solution uses a hybrid architecture combining:

- Real-Time Processing (Azure Event Hub + Azure Functions)
- Batch Processing (Spring Batch + SQL Server)

This architecture supports both modern event-driven systems and legacy systems.

Business Problem

Before CDP implementation:

- Customer information was distributed across multiple systems.
- Duplicate customer records existed.
- No unified customer identity.
- Difficult to perform customer analytics.
- No centralized Customer 360 capability.
- Marketing and servicing teams had fragmented customer visibility.

The CDP addresses these challenges by creating a centralized customer repository.

Business Objectives

Primary objectives:

- Customer 360
- Single Source of Truth
- Identity Resolution
- Journey Tracking
- Customer Analytics
- Personalization
- Cross-Sell and Upsell
- Regulatory Reporting

High-Level Architecture

Source Systems

- Health Insurance
- Travel Insurance
- Motor Insurance
- Term Insurance
- CRM
- Mobile App
- Web Portal
- Partner Systems

Ingestion Layer

Real-Time

- Azure Event Hub
- Azure Functions

Batch

- Spring Batch
- Stored Procedures

Identity Resolution Layer

Customer Profiles

- Prospect Profile
- Golden Profile

Customer 360 APIs

Consumers

- CRM
- Marketing

- Analytics
- Customer Service

Real-Time Processing Architecture

Business Events

Examples:

- Quote Created
- Proposal Submitted
- KYC Completed
- Payment Success
- Policy Issued
- Customer Login

Processing Flow

Application

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Azure Event Hub

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Azure Functions

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Validation

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Identity Resolution

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CDP Database

Benefits

- Near real-time updates
- Immediate customer visibility
- Event-driven architecture

Azure Components

Azure Event Hub

Purpose

Enterprise event ingestion platform.

Responsibilities

- Event streaming
- Partition management
- Event retention
- High throughput

Comparable To

Kafka as a managed Azure service.

Azure Functions

Purpose

Serverless event processing.

Responsibilities

- Event validation
- Transformation
- Customer matching
- Journey updates

Benefits

- Auto scaling
- No infrastructure management

Azure Storage Account

Used for:

- Raw event storage
- Replay support
- Audit requirements

Azure Key Vault

Stores:

- Secrets
- API Keys
- Database Credentials

Azure Monitor

Provides:

- Operational monitoring
- Alerting
- Metrics

Application Insights

Provides:

- Tracing
- Dependency monitoring
- Exception tracking

Batch Processing Architecture

Batch jobs synchronize:

- Quotes
- Proposals
- Members
- KYC
- Payments
- Policies

Flow

Stored Procedure

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Reader

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Processor

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Writer

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CDP Database

Framework

Spring Batch

Features

- Restartability
- Chunk Processing
- Parallel Execution
- Audit Logging

Identity Resolution Engine

Most critical CDP component.

Matching Hierarchy

1. Customer ID
2. PAN
3. Mobile
4. Email
5. Composite Rules

Purpose

Identify whether incoming data belongs to:

- Existing Customer
- Existing Prospect
- New Customer

Prospect Profile

Represents a potential customer before policy issuance.

Contains

- Contact Information
- Demographics
- Quote History
- Proposal History
- Journey Participation

Golden Customer Profile

Represents enterprise customer identity.

Contains

- Customer ID
- Master Contact Details
- Customer Preferences
- Products Owned
- Relationships

Purpose

Single Source of Truth.

Customer Journey Model

Quote

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Proposal

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KYC

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Payment

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Policy

Benefits

- Funnel Analysis
- Drop-off Analysis
- Conversion Tracking

Customer 360 View

Provides:

- Profile
- Quotes

- Proposals
- KYC
- Payments
- Policies
- Interaction History

API

GET /customer/{customerId}/360

Data Model

Core Entities

- ProspectProfile
- GoldenProfile
- CustomerRelationship
- CustomerContact

Journey Entities

- QuoteData
- ProposalData
- MemberData
- KYCData
- PaymentData
- PolicyData

Reference Data

- ProductMaster
- LOBMaster
- ChannelMaster
- OperatorMaster

API Strategy

Profile APIs

- GET /profiles/{id}
- GET /profiles/search
- GET /profiles/customer/{customerId}

Journey APIs

- GET /quotes
- GET /proposals
- GET /payments
- GET /kyc

Customer APIs

- GET /customer/{customerId}/360
- GET /customer/{customerId}/timeline

Security Architecture

Authentication

OAuth2

Authorization

Role-Based Access Control (RBAC)

Encryption

TLS

PII Protection

- PAN
- Email
- Mobile

Audit Logging enabled.

Monitoring & Observability

Tools

- Azure Monitor
- Application Insights
- SQL Monitoring
- Spring Batch Monitoring

Metrics

- Event Throughput
- Processing Latency
- Failed Events

- Batch Job Success Rate
- API Response Time

Scalability Strategy

Real-Time

Event Hub Partitions

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Multiple Function Instances

Batch

Spring Batch Partitioning

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Parallel Workers

Database

- Index Optimization
- Partitioned Tables
- Read Replicas

Architecture Decisions & Trade-Offs

Why Event Hub Instead of Kafka?

Event Hub chosen because:

- Azure Native
- Fully Managed
- Lower Operational Cost
- Easier Monitoring

Kafka would be preferable in multi-cloud environments.

Why Batch and Real-Time Together?

Real-Time

- Customer interactions
- Payments
- Quotes

Batch

- Legacy systems
- Historical loads

Hybrid architecture ensures complete coverage.

Why SQL Server Instead of NoSQL?

Insurance data is highly relational.

Requirements:

- ACID Transactions
- Reporting
- Referential Integrity

SQL Server provides better support.

Why Golden Profile?

Without Golden Profile:

- Duplicate Customers
- Inconsistent Reporting
- Poor Analytics

Golden Profile creates a Single Source of Truth.

Why Spring Batch Instead of Data Factory?

Advantages:

- Complex business logic
- Better Java integration
- Fine-grained control
- Restartability

Why API Access Instead of Direct Database Access?

Benefits:

- Security
- Governance
- Versioning
- Scalability

- Encapsulation

Principle

Source Systems Own Data

CDP Owns Customer View

Architecture Interview Questions & Answers

Key Topics:

- Event Hub vs Kafka
- Azure Functions
- Idempotency
- Duplicate Prevention
- Event Ordering
- Event Replay
- Golden Profile Creation
- Mobile Number Changes
- Data Merging
- Customer 360
- Prospect vs Golden Profile
- Hybrid Architecture
- Consistency Management
- Scalability
- Monitoring
- Security
- Identity Resolution Challenges
- Future Enhancements

Future Roadmap

Phase 1

Prospect Consolidation

Phase 2

Golden Profile

Phase 3

Customer 360 APIs

Phase 4

CDC Integration

Phase 5

Real-Time Profile Updates

Phase 6

AI Identity Resolution

Phase 7

Next Best Action Engine

Phase 8

Customer Segmentation Engine

Interview Closing Statement

"Our CDP is a hybrid architecture that combines Azure Event Hub and Azure Functions for real-time customer events with Spring Batch ETL processing for legacy and bulk data synchronization. The platform performs identity resolution, builds Prospect and Golden Customer Profiles, tracks complete customer journeys, and exposes Customer 360 APIs. The objective is to create a trusted enterprise customer repository that supports analytics, personalization, operational systems, and future AI-driven customer engagement."